

Introduction

See also: [Slide Presentation](#) and [GitHub](#)

In this report, we explore the behavior of professional baseball players in the United States to better understand the characteristics that make competitive players stand out. We first look at player loyalty in general to understand how common or uncommon it is to switch teams during a career. We then extend loyalty and compare it with batting average to determine if team switching is correlated to skill. Then we compare loyalty with pay to determine if team switching is correlated to higher pay. Finally, we look at trends in player salaries against age. These findings can inform investment decisions made by baseball teams. Although we do not suggest that these findings are universally applicable, some findings will be interesting to non-athletes who are concerned with remaining competitive in their respective field.

Dataset

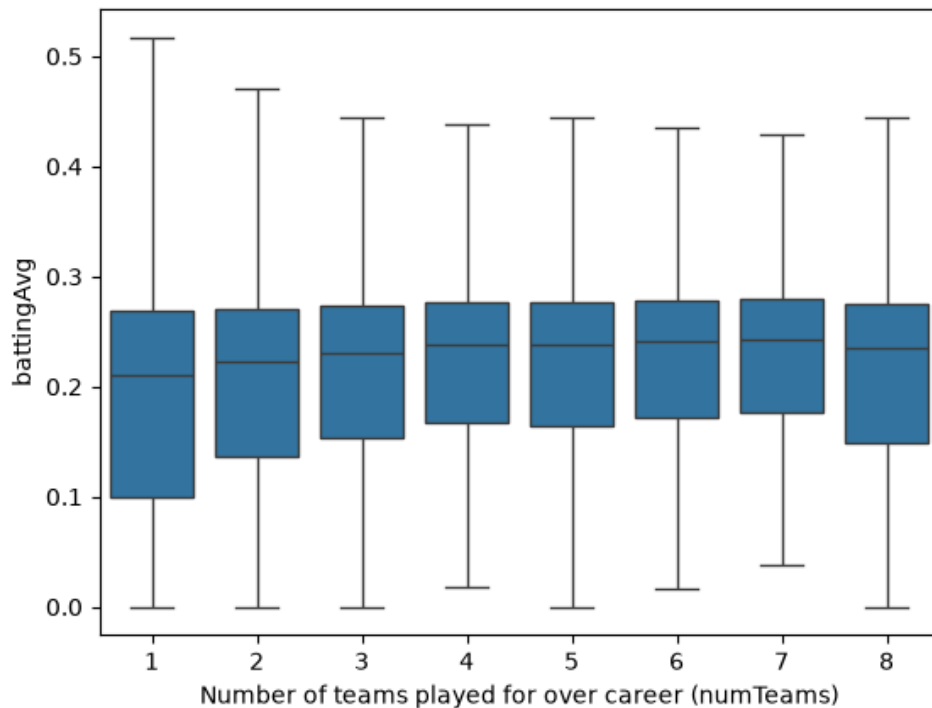
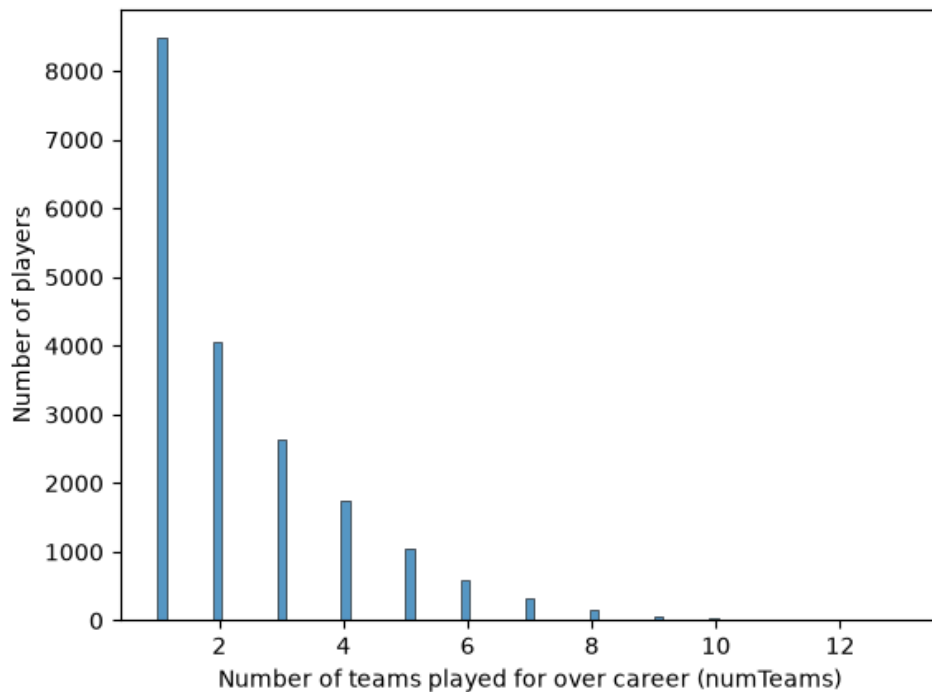
Our analysis uses The Lahman Baseball Database dated January 24, 2015, which describes itself as “the largest and most accurate source for baseball statistics available anywhere.” The Lahman database is a widely-recognized database with a large community backing and is comprehensive for our use case. It contains information about player performance, age, appearances, team affiliations, etc. While extensive, it is small enough (~33 MB) to fit in memory and is structured in a relational format that makes computation straightforward. The database is open source and publicly accessible, making it easy to replicate and validate our results.

Analysis Technique

Much of our analysis is visual, with some statistical models used to analytically verify details that are graphically less obvious. We use histograms and boxplots to look at the distribution of univariate data such as the number of teams a player plays for over his career (career teams). We use scatterplots to look at bivariate distributions, such as when comparing salary with other variables. We use a barplot to compare average career salary to the number of career teams and to compare the average player salary of the highest paying teams because of the discrete (categorical) nature of number of career teams and team name. We use a two-sample t-test to compare the batting average distribution of players who played for one team versus those who played for four teams to confirm the relationship vaguely present in the graphs. We use a Pearson correlation (standard linear regression correlation statistic) matrix to compare the relationship between career teams, batting average, and career salary, but we excluded these results from the report because we did not find them interesting.

Results

In general, it is very common for players to remain with the same team for their entire career (~44%). 65% of players change teams once or less over the course of their career. However, the number of teams a player plays for over the course of their career is positively correlated with

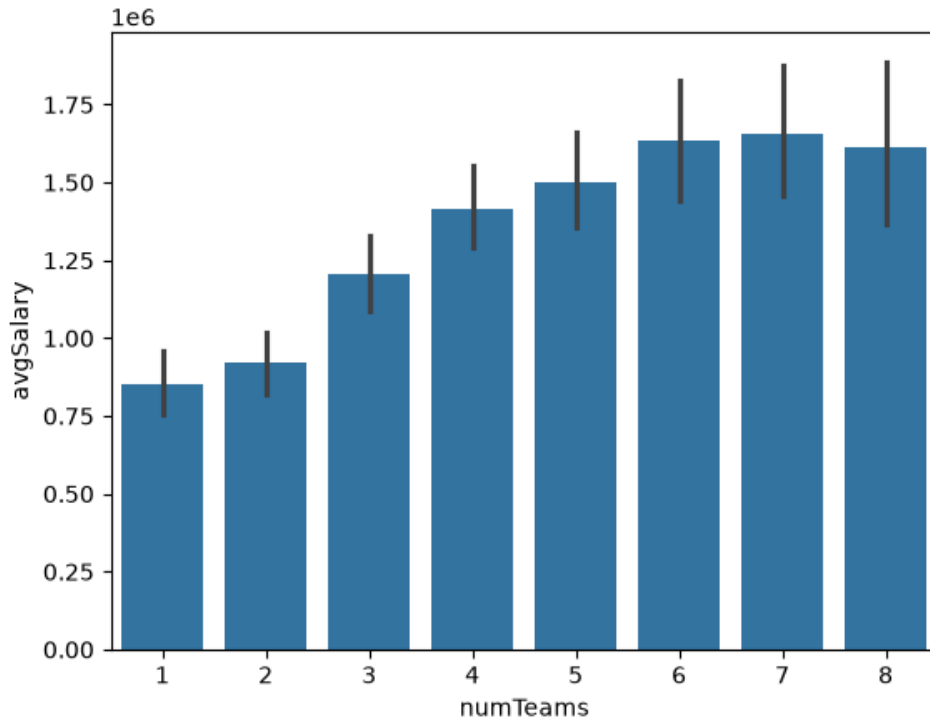


skill level, as measured by batting average. The box plot below shows that this correlation is strongest up to and including those who have played for four teams over their career. Switching teams any more than three times (four teams total) does not appear to be correlated with better batting averages. A two-sample t-test of batting averages of players playing for two teams versus three teams yields $p=2.9e-8$, showing a statistically significant difference between the batting averages of both athlete groups. While we lack the data to suggest causation between the number of career teams and batting average, this correlation might make players wishing to remain competitive more open to considering team switches.

Players wishing to maximize pay, however, might consider the

opposite: building tenure with their current team. Although average salary increases with the number of career teams, we consider this correlation misleading because players with more team switches generally have longer careers.

To illustrate the advantage of building tenure on a team, we compute the average salary raise for athletes who remained with the same team year-over-year and those who switched teams in the



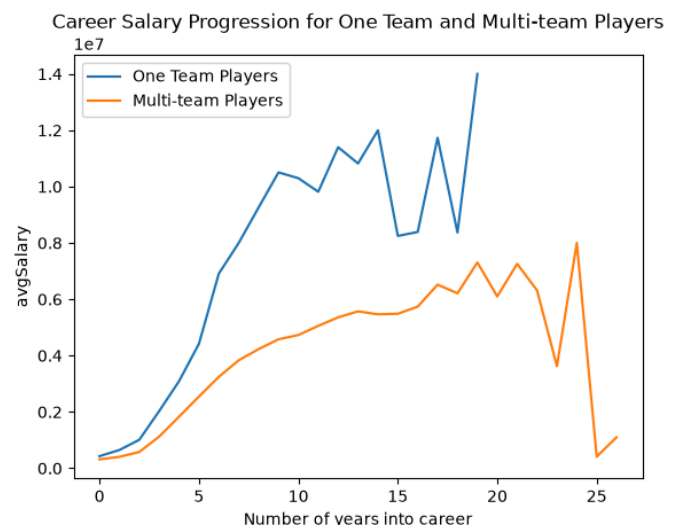
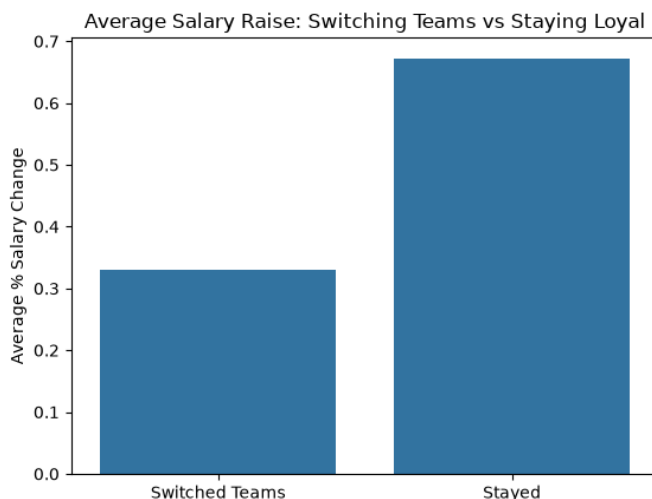
offseason (those switching teams during the year were excluded because salaries are prorated in this case). We also graph the average salary versus career year to visualize career salary progression for one team players and multi-team players.

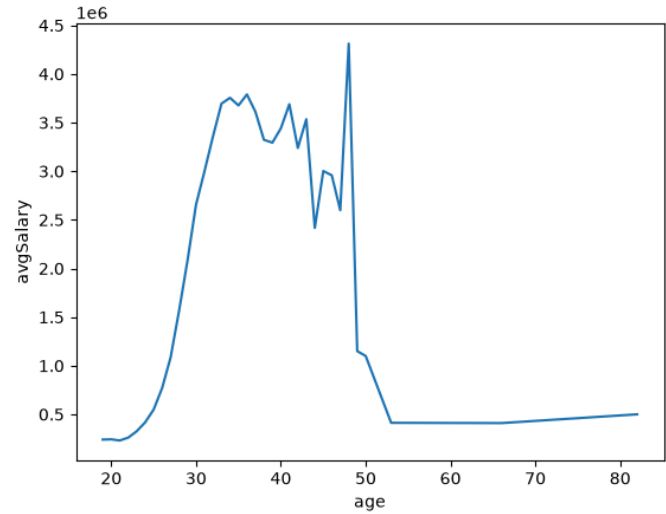
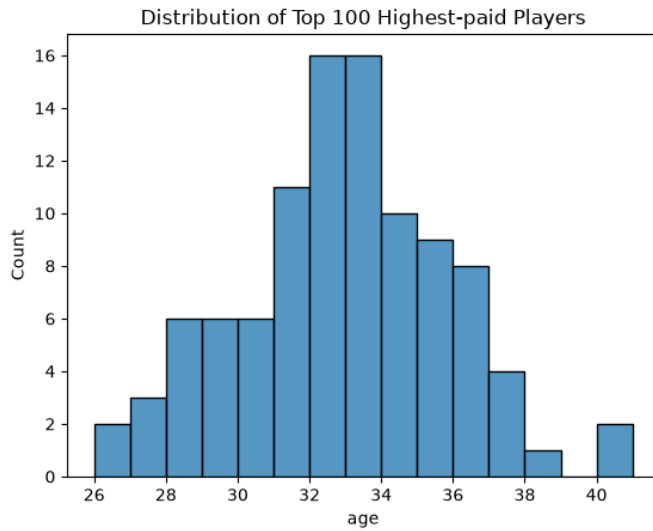
Finally, we consider the impact of age on a player's pay. The histogram of the top 100 highest paid players suggests that the 32–34 range is ideal from a

salary perspective. The line graph of average salary over age shows a similar general trend.

Technical

For the most part, we use standard pandas/numpy operations with seaborn graphs as discussed in class. We compute the two-sample t-test using the Scipy stats library. Our most complicated analyses were computing average salary raise and career salary progression. The former uses the `pct_change` and `shift` pandas operations on a table sorted by year to track year-over-year changes in salary and team. The latter computes rookie years and career duration for each row in the salary table and averages salaries grouped by career duration. We experimented with the use of a Pearson correlation matrix using the





pandas `corr` function and graphing the results using a seaborn heatmap, although we did not find these results meaningful.